

Report Mate — Business Proposal

Show Me Your Agents Hackathon (NUS-ISS, Public Category)

Team Simpsons — Arrav · Aaron · Ngiam · Letao

2026

Contents

1. Executive summary	2
2. The problem	2
3. The solution	2
3.1 What makes it different	2
4. Who it is for	3
5. Value proposition	3
6. How it works (at a glance)	3
7. Responsible and practical AI	3
8. Current status	4
9. Roadmap	4
10. Team	4

1. Executive summary

Report Mate is a mobile-first application that helps field-service technicians produce accurate, consistent customer visit reports in minutes instead of at the end of a long day. Technicians describe what happened in plain language — typed or dictated — and an AI agent fills the appropriate report template, drawing on the customer’s job history and a real parts catalog, and flagging anything it is not sure about. A human always reviews and approves before the report is submitted.

The result: faster reporting, more complete records, and a documentation process that fits the reality of field work rather than fighting it. Report Mate keeps the technician firmly in control while removing the tedious part of the job.

2. The problem

Field technicians — HVAC, electrical, facilities, appliance repair — document work performed, parts used, customer observations, and follow-up recommendations by hand. This happens under poor conditions: on a phone, in a vehicle, in bright sun, and often after already driving to the next job.

The consequences are familiar to any service business:

- **Slow.** Paperwork eats billable time and stretches the working day.
- **Inconsistent.** Reports vary by technician; important details are missed.
- **Incomplete.** Parts and follow-ups are forgotten, which affects invoicing and repeat visits.
- **Hard to search later.** Inconsistent, hand-written records are difficult for dispatchers to review or audit.

3. The solution

Report Mate is built around a simple loop:

1. A dispatcher/admin builds a report **template** from field blocks (text, number, select, checklist, photo, signature) using a drag-and-drop builder.
2. A technician picks the right template for a job.
3. They either fill it in by hand, or hand an **AI agent** a rough account of the visit. The agent fills the blanks, matches mentioned parts to the real catalog, pulls in the customer’s past visits, and flags anything it cannot confidently complete — rather than guessing.
4. The technician **reviews and edits** the draft. Nothing is submitted without a human check.
5. Dispatchers see a central history of jobs and reports and manage templates over time.

3.1 What makes it different

- **The AI asks instead of guessing.** In conversational mode the agent asks the technician a clarifying question (for example, a meter reading it needs) rather than inventing a value. This is what keeps reports trustworthy.
- **Human-in-the-loop by design.** The agent drafts; the technician is always the author of record. This matters for accountability, warranty, and compliance.
- **It still works when the AI does not.** If the AI service is unavailable, the same template can be completed by hand. The business is never blocked by an outage.
- **Voice input for the field.** Technicians can dictate the account hands-free, which suits the working environment.

4. Who it is for

- **Field technicians** — the primary users, who want to finish reports quickly and correctly without fighting a form on a small screen.
- **Dispatchers / service managers** — who need consistent, complete records, a central history of jobs, and control over report templates.
- **Service businesses** — HVAC, electrical, plumbing, facilities management, and appliance repair firms that run recurring customer visits.

5. Value proposition

Stakeholder	Value delivered
Technician	Less paperwork, faster job turnaround, hands-free voice input
Dispatcher	Consistent, complete, searchable reports; template control
Business	Better invoicing accuracy, fewer missed follow-ups, auditable records
Customer	Clear, professional documentation of work performed

The core benefit is **time returned to billable work** combined with **higher quality records** — the two things a service business cares about most.

6. How it works (at a glance)

Report Mate is a web application that runs on any phone, tablet, or laptop browser. Dispatchers configure templates once; technicians reuse them on every job. The AI agent runs entirely on the secure backend — the technician simply types or speaks, reviews the draft, and submits. Reports can be exported for records and invoicing.

The AI is deliberately scoped: it fills the fields a template defines and flags uncertainty. It does not free-write reports from nothing, which is what keeps the output predictable and safe to sign off on.

7. Responsible and practical AI

Report Mate treats AI as an assistant, not an authority:

- **The human approves every report.** The agent cannot submit on its own.
- **The AI is transparent about uncertainty.** Fields it could not fill are flagged for human attention.
- **Data stays protected.** The AI service is only ever contacted from the secure backend; credentials are never exposed to the browser.
- **Cost is controlled.** Each AI conversation is bounded by turn and question limits so usage stays predictable.

8. Current status

Report Mate is a working product, not a mockup. The template builder, manual and AI-assisted fill paths, human review, dashboard, and report export are implemented, tested, and deployed to a live cloud environment. Both one-shot and conversational AI drafting work end to end against a live AI gateway, including real parts matching and job-history context, with voice dictation available in the chat interface.

9. Roadmap

Near-term opportunities that build on the working foundation:

- **Offline drafting with sync on reconnect** — technicians frequently lose signal on site; drafting offline and syncing later removes a real pain point.
- **Template versioning** — track and roll back changes to report templates over time.
- **Deeper analytics** — surface trends across jobs, parts usage, and technicians for service managers.
- **Attachment storage at scale** — move photos and signatures to object storage for larger deployments.

10. Team

Report Mate was built by **Team Simpsons** for the NUS-ISS “Show Me Your Agents” hackathon (Public Category): **Arrav, Aaron, Ngiam, and Letao** — covering the AI agent, backend services, template and report experience, and deployment.